

Revitalizing Traditional Animation

Pre-Composite Frame Interpolation as a Production Catalyst

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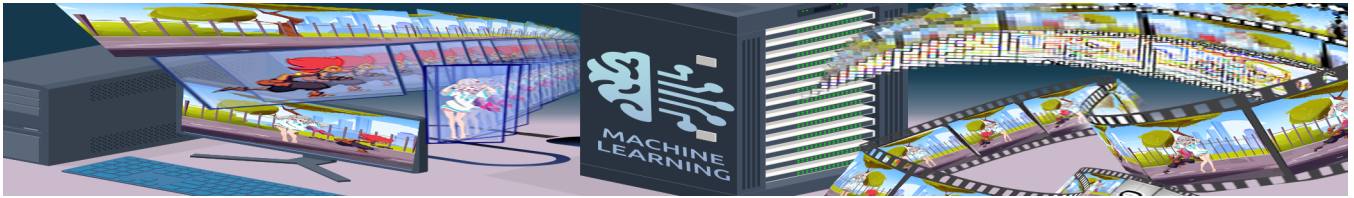


Figure 1: A single image narrative of our method of interpolating elements of animations before they are composited.

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1 INTRODUCTION

For over a century, audiences of all ages have been entertained by the captivating art of animation. Since Toy Story's groundbreaking release in 1995, animation has increasingly shifted to being produced in 3D computer generated and 2D vector graphics, marginalizing traditional hand-drawn animation, especially in US feature films. While digital tools can replicate traditional processes, there's a unique opportunity for computational innovation to collaborate with skilled artists to revitalize this art form. This paper examines the gap between computer scientists' current in-betweening techniques, like video frame interpolation, and proposes an alternative method customized for traditional animation's workflow, aiming for greater precision and a novel approach.

2 IN-BETWEENING

In simple terms, in-betweening involves creating new images from two input images to represent intermediate points. In traditional animation, skilled artists meticulously craft thousands of frames consisting of layers for elements like backgrounds, foregrounds, characters, and objects, both stationary and animated. Lead animators define keyframes that establish core movements, which in-between artists then fill in to bring characters to life with grace.

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Frame Interpolation (FI) is a computational process where two images generate a new image representing a possible "in-between" state. This technique can simulate motion between original images and, when applied to photos or videos, be used to upscale frame rates or create slow-motion effects. FI is a significant topic in computer vision, employing deep convolutional neural networks and depth-aware flow projection techniques to enhance visual processing [Bao et al. 2021; Li et al. 2022; Niklaus et al. 2017].

3 PRE-COMPOSITE FRAME INTERPOLATED IN-BETWEENING

Frame interpolation involves computationally generating intermediate frames between existing frames in photos or videos, just like in-betweening. Researchers have applied this technique to traditional animation by using captures from films to recreate specific in-between frames for comparison with the original. However, using these captures includes post-production effects, which can be a limitation. Previous literature (e.g., [Bao et al. 2021; Chen and Zwicker 2022; Li et al. 2022; Shen et al. 2022; Siyao et al. 2021]) may have based their processes on data from the wrong stage in the animation production cycle.

Current frame interpolation systems also evaluate entire frames holistically, unlike traditional hand-drawn animation where multiple layers of images create a single frame. To mimic the animation in-betweening process more accurately, the focus should be on processing the individual keyframes of each layer separately. This approach isolates elements within the frame, reducing conflicts between elements on different layers. Consequently, the interpolation process can generate more precise in-between frames that require fewer corrections for issues like image blurring, color smearing, and element deformation caused by proximity to other elements. Once each layer's keyframes are interpolated, they can be finalized through compositing.

3.1 METHOD AND IMPLEMENTATION

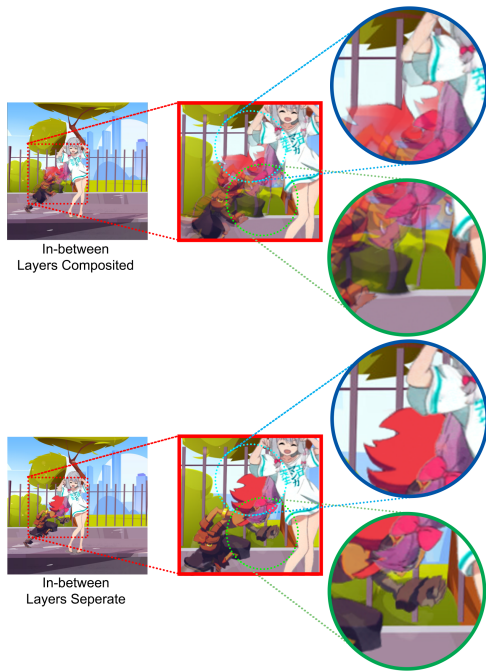
We used deconstructed animated gifs to create simple keyframes for testing our interpolation methodology. Each keyframe includes



Figure 2: Elements of the animation used for our pre-composite testing.

a static background (Figure 2a)¹, a running character (Figure 2b)², and a dancing character (Figure 2c)³. The runner passes behind the dancer on a park sidewalk from left to right. One set of keyframes combines all elements per frame, while another set consists of 15 keyframes per element. Using the Flowframes application [N00MKRAD 2020], we expanded each set from 15 to 30 frames using RIFE (Real-Time Intermediate Flow Estimation) [Huang et al. 2022], a neural network for intermediate flow estimation between consecutive RGB frames. This resulted in 30 frames based on composited keyframes and interpolated in-betweens, and another 30 frames where each element’s keyframes and in-betweens were composited after interpolation.

3.2 Results



(a) (Top) Results based on interpolating composited keyframes.

(b) (Bottom) Results based on interpolating each layer’s keyframes individually.

Figure 3: Methodology Test Results

¹“Summer City Skyline Urban View Fence” [Pauliukevich 2022]

²“Young Ganondorf” [Dupouy 2020]

³“Excerpt from ‘Eromanga Sensei’” [Takeshita and Takahashi 2017]

Figure 3a shows an interpolated in-between using composited element keyframes, while Figure 3b demonstrates individual interpolation of each element’s keyframes before compositing. In Figure 3a, the interpolation algorithm considers the entire frame simultaneously, potentially distorting the overall image. This results in visible distortions such as irregular bending of the park fence, a semi-transparent effect on the runner, and ghosting on the dancer’s hair and arm. This is due to their proximity to other elements in the frame. In contrast, Figure 3b maintains the integrity of individual elements: the park fence remains straight, the runner passes clearly behind the dancer, and the dancer’s hair and arm retain sharp definition. By interpolating each element separately, the algorithm focuses solely on the visible layers being processed, yielding cleaner and more precise results between adjacent elements.

4 FUTURE WORK

We seek partnerships with animation schools, studios, and independent animators to test interpolation systems using our methodology. Access to animation data is crucial for evaluating our approach, which initially lacked in-betweens due to low frame counts. Our goal is to observe how materials are interpolated and analyze differences in visuals, PSNR, and SSIM compared to pre-established in-betweens.

Our vision is to sustain traditional animation’s vibrancy in a changing creative and computational landscape. Implementing frame interpolation streamlines processes, reducing time, costs, and staffing needs while preserving artists’ roles. We aim to empower individual creators to achieve team-level output, enable smaller studios to expand their impact, and lower overall production expenses. Our approach supports artists in captivating audiences, fostering creativity, and bringing dreams to life for future generations.

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